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SUPERSEDING
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FEDERAL SPECIFICATION

CLOTH, COTTON, DUCK, UNBLEACHED, PLIED-YARNS, (ARMY AND NUMBERED)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements of grey (unbleached) cotton duck cloth.

1.2 Classification.

1.2.1 *Types.* The cloth, cotton, duck, plied-yarns, (Army and Numbered) shall be of the following types as specified (see 6.2) :

Type I—Numbered duck.

Type II—(see 6.4).

Type III—Army duck.

Type IV—(see 6.4).

2. APPLICABLE SPECIFICATIONS, STANDARDS AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

Federal Specifications:

CCC–T–191—Textile Test Methods.

PPP–P–51 — Packaging, Packing and Marking of Textile Fabrics (Woolens, Worsteds, Cotton, Silk and Synthetics).

Federal Standards:

Fed. Std. No. 4—Glossary of Fabric Imperfections.

Fed. Std. No. 102—Preservation, Packaging and Packing Levels.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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TABLE I.—*Physical requirements—type I, numbered duck*

Number of duck, commercial designation	Weight, per square yard, minimum	Yarn ply, minimum		Yarns per inch, minimum		Breaking strength, minimum		Air permeability, maximum
		Warp	Filling	Warp	Filling	Warp	Filling	
	Ounces					*Pounds*	*Pounds*	*Ft.³/min./ft.²*
Hard texture								
12/0	48.00	9	9	20	12	800	430	--
8/0	40.00	8	14	26.5	12.5	750	600	--
2/0	31.90	5	7	24	16	465	435	--
1/0	30.31	5	6	24	16	450	405	--
1	28.71	5	5	26	19	440	370	2
2	27.12	5	4	26	19	420	345	--
3	25.53	4	5	29	20	390	330	--
4	23.93	4	4	31	22	375	300	2
5	22.33	3	4	35	23	345	285	--
6	20.74	3	3	35	25	335	250	2
8	17.55	3	3	43	25	285	210	2
10	14.35	3	3	45	27	245	160	4
11	12.75	3	2	45	34	240	140	--
12	11.16	2	2	45	35	195	120	4
Medium texture								
1	28.71	5	5	26	15	425	345	--
2	27.12	5	4	26	15	410	320	--
3	25.53	4	5	29	15	370	315	--
4	23.93	4	4	29	17	350	290	--
5	22.33	3	4	34	18	315	285	--
6	20.74	3	3	34	18	305	250	--

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TABLE II.—*Physical requirements—type III, Army duck*

Weight per square yard, minimum	Yarn ply, minimum		Yarns per inch, minimum		Breaking strength, minimum		Air permeability, maximum
	Warp	Filling	Warp	Filling	Warp	Filling	
Ounces					*Pounds*	*Pounds*	*Ft.³/min./ft.²*
8.25	2	2	60	58	125	120	--
9.85	2	2	52	40	160	110	--
12.29	3	2	44	34	210	130	4
14.77	3	3	44	28	235	175	--
15.90	3	3	45	34	245	200	--
18.48	4	4	42	28	315	200	--

2.2 Other publications. The following publication forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Rules and Regulations under the Textile Fiber Products Identification Act effective March 3, 1960.

(Copies may be obtained free of charge from the Federal Trade Commission, Washington 25, D. C.)

3. REQUIREMENTS

3.1 Material.

3.1.1 *Yarn.* Yarns shall be made from cotton which has been cleaned, carded, drawn, evenly spun, and twisted into plied yarns.

3.2 Physical requirements. Cloth covered by this specification shall be as specified (see 6.2) and shall conform to the requirements specified in tables I and II when tested as specified in 4.4.

3.2.1 *Width.* The width of the cloth, shall be as specified (see 6.2). The following tolerances will be permitted:

Width of duck (inches)	*Tolerance (inch)*
40 or less	+ ¼ — ¼
Over 40 to 51 inclusive	+ ⅜ — ¼
Over 51 to 81 inclusive	+ ⅜ — ⅜
Over 81	+ ¾ — ⅜

3.2.2 *Weave.* The weave shall be plain, one up, one down.

3.3 Nonfibrous material content. Cloth shall contain not more than 2.5 percent starch and protein material and not more than 5.0 percent total size, finishing, and nonfibrous materials including starch and protein when tested as specified in 4.4. The use of finishing or loading materials to increase the weight or breaking strength is prohibited and the contractor shall furnish a certificate of compliance with respect to this requirement.

3.4 Length.

3.4.1 *Put-up.* The cloth shall be put up in rolls in accordance with Federal Specification PPP–P–51. Pieces 80 yards or more in length shall be delivered. The pieces shall be

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sewn together with a butted-type seam on all duck No. 10 and lighter in weight, using approximately 10/3 cord cotton thread, and have approximately 9 stitches per inch. The stitching shall be placed not closer than ½ inch from the cut or raw edge of the cloth. On cloth heavier in weight than No. 10, the pieces will be lapped and not sewn together. The number of pieces in the roll will be indicated by stencil marking on the last 6 inches of the roll on the outer surface (see 6.3).

3.4.1.1 *Short length.* Pieces less than 80 yards will be acceptable up to 10 percent of contract quantity, provided no piece is less than 25 yards in length, and not more than two pieces less than 80 yards are included in any one roll, except where otherwise specified in the contract (see 6.3).

3.5 Identification. Each roll of cloth shall be marked for identification in accordance with Federal Specification PPP–P–51.

3.6 Fiber identification. Each piece or roll of cloth shall be labeled, ticketed or invoiced for fiber content in accordance with the Rules and Regulations of the Textile Fiber Products Identification Act effective March 3, 1960.

3.7 Workmanship. The cotton duck cloth shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the levels set by the applicable acceptable quality levels.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Contractor's inspection responsibility. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection for acceptance. Except when otherwise specified by civil government agencies, inspection shall be in accordance with the provisions of Military Standard MIL–STD–105, except where otherwise indicated.

4.2.1 *Examination of the end item.* The cotton duck shall be examined to determine conformance with the requirements of this specification. Defects found during this examination shall be classified in accordance with 4.2.1.1, 4.2.1.2, and 4.2.1.3.

4.2.1.1 *Yard by yard examination.* All defects listed in table III shall be counted, regardless of their proximity to one another, except where two or more defects represent a single local condition of the cloth, in which case only the more serious defect shall be counted. A continuous defect shall be counted as one defect for each warp-wise yard or fraction thereof in which it occurs. The unit of product for this examination shall be 1 linear yard. The acceptable quality level for this examination shall be as follows:

1. Specified width 51 inches or less—0.4 major defects and 2.5 total defects per 100 units (yards).
2. Specified width over 51 inches—0.65 major defects and 4.0 total defects per 100 units (yards).

Unless otherwise specified, the inspection level shall be level II of Military Standard MIL–STD–105. The lot size shall be expressed in terms of 1 yard each. An approximately equal number of yards shall be examined from each roll in the sample. The number of rolls from which the sample is to be selected shall be in accordance with table IV.

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TABLE III.—*Defects*

Glossary No. Fed. Std. No. 4	Defect	Classification	
		Major	Minor
55, 56, 90, 91, 92	Hole, cut, tear or break (including selvage) in which more than three yarns are ruptured in adjoining points	X	
78	Smash—any	X	
79, 89	Abrasion mark, bruise, tender or weak spot—any	X	
11, 33	Broken or missing yarns—		
	Three or more contiguous regardless of length	X	
	Two contiguous, 2 inches or more missing	X	
	Two contiguous, less than 2 inches or single 2 inches or more missing		X
9	Fine or light filling bar, light place—clearly visible[1]	X	
2	Coarse filling bar or heavy place—clearly visible[1] and exceeding 2 linear inches		X
5, 27, 29, 77	Slough off, jerked in filling, slubs, gouts, knots—more than three times the thickness of the normal yarns		X
9, 12	Open place or crack—clearly visible[1]	X	
13, 34	Floats—multiple, ½ inch or more in combined warp and filling directions, or harness skips	X	
	Multiple less than ½ inch in combined warp and filling directions, or single 2 inches or more in length		X
41	Reed mark—resulting in a clear warpwise separation markedly wider than normal		X
40	Wrong draw—½ yard or more		X
49, 50	Slack or tight selvage resulting in pronounced slackness or waviness in cloth		X
51	Scalloped selvage—indentations exceeding ½ inch		X
48	Curled, wrinkled or folded selvage—½ yard or more		X
53, 54	Stringy, beaded, corded, loopy or rough selvages—½ yard or more, resulting in raised places along the selvage edge two or more times the normal fabric thickness, ½ yard or more		X
36	Mixed ends, clearly visible[1] and causing definite appearance of bars (streaks)		X
30	Untrimmed yarn tails—single, more than ¼ inch in length		X
74, 75, 76	Spot, stain or streak (from oil or grease)—uncleaned, covering more than 1 inch in combined directions	X	

[1] Clearly visible at normal inspection distance (approximately 3 feet).

TABLE IV.—*Sample size*

Lot size in yards	Sample size in rolls	Maximum number of defects acceptable in sample
Up to 1300[1]	3	0
1,301 to 3,200	5	0
3,201 to 8,000	7	0
8,001 to 22,000	10	0
22,001 to 110,000	15	0
110,001 and over	25	1

[1] If the lot contains fewer than three rolls each roll in the lot shall be examined.

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4.2.1.2 *Overall examination.* The unit of product for this examination shall be the entire roll of cloth. The sample size and acceptance number shall be shown in table IV. The lot size shall be expressed in units of one linear yard.

Defects

Baggy, ridgy or wavy cloth—clearly visible.[1]
Trash, motes, seed coat fragments, overall uncleanness—clearly visible.[1]
Rancid or otherwise objectionable odor.
Any selvage defects as listed in table III that occur continuously or intermittently over more than half the yardage examined.
Uneven weaving throughout.
Width not within specified tolerance.

[1] Clearly visible at normal inspection distance (approximately 3 feet).

4.2.1.3 *Length examination.*

4.2.1.3.1 *Individual rolls.* Each defect listed below shall be counted not more than once in each roll examined. The unit of product shall be one roll. The lot size shall be expressed in units of one yard each. The sample size and acceptance number shall be as shown in table IV.

Defects

Any gross length found to be more than 2 yards below the gross length marked on the roll ticket.
Any piece less than 80 yards in length.
For duck heavier in weight than No. 10, pieces sewn instead of lapped.
For No. 10 duck and lighter in weight:
(*a*) Pieces not sewn together with a butted type seam.
(*b*) Approximately 10/3 cord cotton thread not used.
(*c*) Approximately 9 stitches per inch not used in seam.
(*d*) Stitching placed closer than ½ inch from the cut or raw edge of duck.
Number of pieces in roll not stencilled on last 6 inches of roll on outer surface.

4.2.1.3.2 *Total yardage in sample.* The lot shall be unacceptable if the total of the actual gross length of rolls in the sample is less than the total of the gross lengths marked on piece tickets.

4.3 Examination of preparation for delivery requirements. An examination shall be made to determine that packaging, packing and marking requirements of section 5 of this specification are complied with. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping roll fully prepared for delivery. The lot size shall be the number of shipping rolls in the inspection lot. The inspection level shall be L–4 and the AQL shall be 4.0 defects per hundred units.

Examine	Defect
Marking	Omitted, incorrect; illegible, of improper size, location, sequence, or method of application.
Material	Any component missing.
	Any component damaged, affecting serviceability.
Workmanship	Inadequate application of components, such as: incomplete closure of rolls, inadequate securing or sealing of wrapping materials, insecure sewing or loose wire ties.
Weight	Gross weight beyond the limits specified.

4.4 Testing of the end item. The methods of testing specified in Federal Specification CCC–T–191 wherever applicable and as listed in table V shall be followed. The physical and chemical values specified in section 3 apply to the average of the number of test specimens selected for a unit of product for test purposes as specified in the applicable test methods. The unit of product for testing shall be 1½ yards full width of the cloth. The lot shall be unacceptable if one or more units of product fail to meet any requirements specified. When the data in the "number of determinations per sample unit" and the "results reported as" columns are not specified in the following table they shall be as required by the referenced test methods. For each characteristic, the sample size shall be as follows:

Lot size in yards	Sample size
Less than 800	2
801 to 22,000	3
22,001 and over	5

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TABLE V.—*Test methods*

Characteristic	Requirement reference	Test method	Number determinations per individual unit of product	Results reported as:
Weight	Table I and II	5041	5	Average of 5 determinations to nearest 0.01 ounce
Weave	3.2.2	Visual	1	Pass or fail
Breaking strength	Table I and II	5100		
Yarn ply	Table I and II	Visual	1	Pass or fail for both warp and filling
Yarns per inch	Table I and II	5050[2]		
Nonfibrous[1] materials	3.3	2611		
Air permeability	Table I and II	5450		

[1] The amount of nonfibrous materials shall be determined with the following formula:

$$\text{Total nonfibrous material, percent} = \frac{O-S}{O} \times 100$$

$$\text{Starch and protein material, percent} = \frac{O-S-W-C}{O} \times 100$$

[2] Except that for type I, 8/0 HT, the results shall be reported to the nearest 0.1 of a yarn.

5. PREPARATION FOR DELIVERY

(Federal Standard No. 102 shall be referred to for definitions and application of the various levels of packaging and packing for civilian agency procurement.)

5.1 Packaging. Packaging shall be level A, B, or C as specified (see 6.2).

5.1.1 *Levels A, B, and C.* The cloth shall be put up on rolls and packaged in accordance with the applicable requirements for class 4 fabrics of Federal Specification PPP–P–51.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 *Levels A, B, and C.* The cloth shall be packed in accordance with the applicable requirements of Federal Specification PPP–P–51.

5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the applicable requirements of Federal Specification PPP–P–51.

6. NOTES

6.1 Intended use. The cloth covered by this specification is primarily intended for use in fabrication of tentage and equipage.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

a. Title, number and date of this specification.
b. Type, number, texture and weight of cloth required (see 1.2 and 3.2).
c. Width of cloth required (see 3.2.1).
d. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
e. Whether special marking is required (see 5.3).
f. Whether short piece will be acceptable (see 6.3).
g. Provisions for short length (see 3.4.-1.1).

6.3 Short pieces. Pieces less than 80 yards in length in excess of the amount specified in 3.4.1.1 and pieces shorter in length than

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25 yards may be presented to the contracting officer for consideration for end uses where short length pieces may be considered.

6.4 Types II and IV have been deleted and will not be required or cited for procurement.

6.5 Transportation description and minimum weights applicable to this commodity are:

Rail:

Cotton piece goods, not otherwise indexed by name, in the original piece.
Carload minimum weight — Any quantity.

Motor:

Cotton piece goods, not otherwise indexed, in the original piece.
Motor volume minimum weight — Any quantity.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded, by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTEREST:
Army—M O Q
Navy—S MC Sh W Y
Air Force—WADD.

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